

Superconductivity: Phenomenological Models

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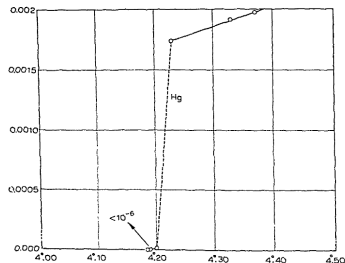
Mathematical Engineering
Quantum Engineering

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Superconductivity

$$\rho = 0$$

$$\chi = -1$$



Perfect conductor

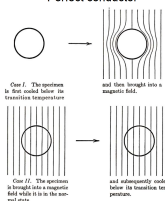


FIG. 3. The behavior expected for a transition into a state of perfect conductivity. The final state would depend on the serial order in which the specimen is brought into the same external conditions.

$$\frac{\partial B}{\partial t} = 0$$

Superconductor

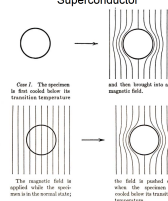
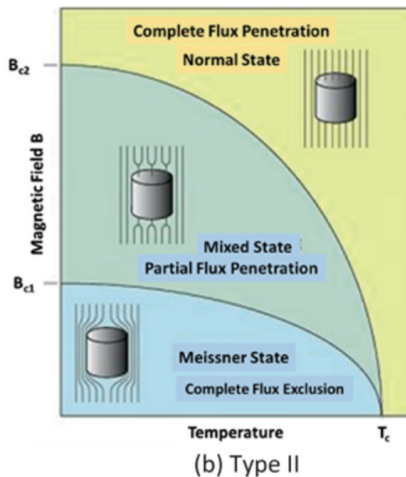
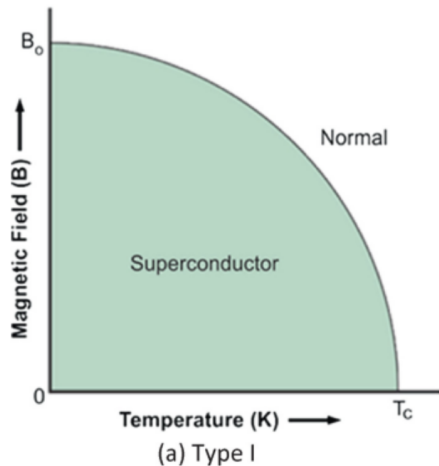


FIG. 4. Case II of Fig. 3 according to Meissner. The superconductor, in contrast to the perfect conductor, has zero magnetic induction independently of the way in which the superconducting state has been reached.

$$B = 0$$

Superconductivity(II)



1. London model for superconductors

London : stationary case (I)

London's equations form a *phenomenological model*: they describe supercurrent j_s and $B = 0$.

Assumed: $\rho = 0$ and uniform behaviour.

First London equation is derived:

$$\begin{cases} F = -eE = m_s a_s \\ j_s = -en_s v_s \end{cases} \implies \frac{dj_s}{dt} = \frac{e^2 n_s}{m} E$$

Meaning: if $E=0$, then $j_s = \text{constant}$ (does not stop).

Second London equation comes applying $\nabla \times E = -\frac{d}{dt}B$ and integrating $\frac{d}{dt} \left(\nabla \times j_s + \frac{e^2 n_s}{m} B \right) = 0$.

Therefore $\left(\nabla \times j_s + \frac{e^2 n_s}{m} B \right) = C \rightarrow C = 0$ to satisfy Meissner effect.

London : stationary case (II)

London equations in the static general case give

$$\nabla \times \mathbf{j}_s = -\frac{1}{\mu_0 \lambda_L^2} \mathbf{B} \quad \text{with} \quad \lambda_L = \sqrt{\frac{m}{\mu_0 n_s e^2}}$$

Taking the curl and using Ampère's law $\nabla \times \mathbf{B} = \mu_0 \mathbf{j}_s$, one gets

$$\nabla \times (\nabla \times \mathbf{B}) = -\frac{1}{\lambda_L^2} \mathbf{B}.$$

Since $\nabla \cdot \mathbf{B} = 0$, it follows

$$\nabla^2 \mathbf{B} = \frac{1}{\lambda_L^2} \mathbf{B}$$

This shows that the magnetic field penetrates only over the characteristic length λ_L , called *London penetration depth*.

London : stationary case (III)

Example For a rectangular superconductor occupying $x > 0$, with external field $B(0) = B_0$ parallel to the surface, the problem becomes one-dimensional:

$$\frac{d^2 B}{dx^2} = \frac{1}{\lambda_L^2} B.$$

with solution

$$B(x) = B_0 e^{-x/\lambda_L}$$

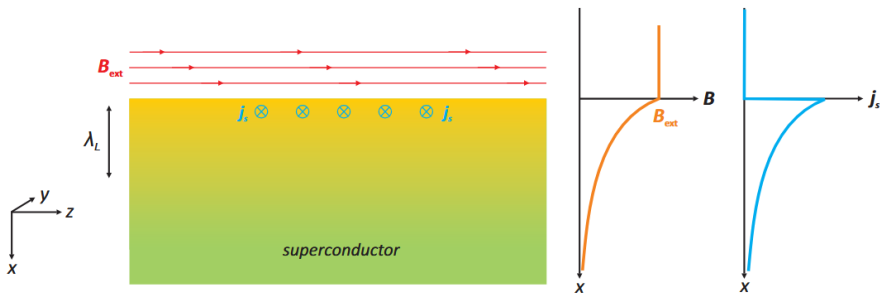
→ exponential decay from the surface into the bulk.

London : stationary case (IV)

Same behaviour for:

$$j_s(x) = \frac{B_0}{\mu_0 \lambda_L} e^{-x/\lambda_L}$$

Therefore both B and j_s are confined near the boundary, within a thickness of order λ_L .



2. Ginzburg-Landau model for superconductors

Landau theory of phase transitions

Landau theory describes a phase transition introducing an *order parameter* η .

Assume: thermodynamic state of the system is the one that **minimizes the free energy**.

Then : phase transition is identified by how the minimum of the free energy changes as temperature varies.

Order parameter : $\eta = 0, T > T_c$ $\eta \neq 0, T < T_c$

Near the transition, the free energy is expanded in powers of η :

$$F = F_n + a(T)\eta^2 + \frac{b}{2}\eta^4 + \dots \quad \text{with} \quad b > 0$$

Landau theory of phase transitions (II)

Let's stop the series at η^4 . The state in which we find the system has $\eta = \eta_{0,1}$ as minimizer of F (as stated):

$$\frac{dF}{d\eta}(\eta_0) = 0$$

$$2a(T)\eta + 2b\eta^3 = 0$$

Solutions: $\eta_0 = 0$ or $\eta_1^2 = -\frac{a(T)}{b(T)}$

η_1 exists (and real) iff $a(T) < 0$.

Therefore : $a(T) = a(T - T_c)$

(We know phase transition occurs at $T=T_c$)

Ginzburg-Landau theory for superconductors

Ginzburg-Landau theory applies Landau's idea to superconductivity.

$$\eta \rightarrow \psi(r) = |\psi(r)|e^{i\theta(r)}$$

Order parameter is a *complex field* where

$$|\psi(r)|^2 = n_s(r)$$

it measures the local density of superconducting electrons.
(According to thermodynamics) in presence of $B=0$, the free energy to be used is the **Gibbs free energy**.

$$G := H - TS$$

Ginzburg–Landau theory for superconductors(II)

Gibbs free energy of a superconductor is

$$G_s = G_n + \int d^3r \left[\alpha |\psi|^2 + \frac{\beta}{2} |\psi|^4 + \frac{1}{2m^*} |(-i\hbar\nabla - q^*A)\psi|^2 + \frac{B^2}{2\mu_0} \right]$$

with $q^* = 2e$, $m^* = 2m_e$.

Meaning:

- **yellow** is the Landau thermodynamic potential, it determines whether the superconducting phase is energetically favorable.
- **purple** describes the kinetic energy of the superconducting condensate.
- **orange** is the magnetic field energy density stored in the electromagnetic field.
- G_n is Gibbs free energy when the system is in the normal metallic phase.

Ginzburg-Landau theory for superconductors(III)

The equilibrium state is obtained minimizing the Gibbs free energy with respect to ψ^* :

$$\frac{\delta G_s}{\delta \psi^*} = 0$$

$$\alpha\psi + \beta|\psi|^2\psi + \frac{1}{2m^*}(-i\hbar\nabla - q^*A)^2\psi = 0$$

This is the **first Ginzburg–Landau equation**.

Ginzburg-Landau theory for superconductors(IV)

(Using Maxwell equation $\nabla \times \mathbf{B} = \mu_0 \mathbf{j}_s$) the **second Ginzburg-Landau equation** is obtained minimizing the Gibbs free energy with respect to the vector potential $\mathbf{A}$:

$$\frac{\delta G_s}{\delta \mathbf{A}} = 0$$

$$\mathbf{j}_s = \frac{q^* \hbar}{2m^* i} (\psi^* \nabla \psi - \psi \nabla \psi^*) - \frac{q^{*2}}{m^*} |\psi|^2 \mathbf{A}$$

→ supercurrent is determined by ψ and $\mathbf{A}$.

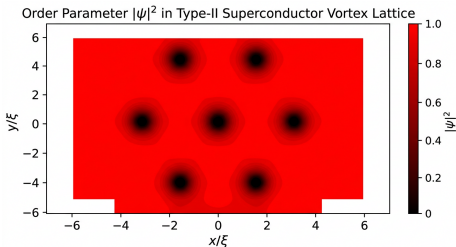
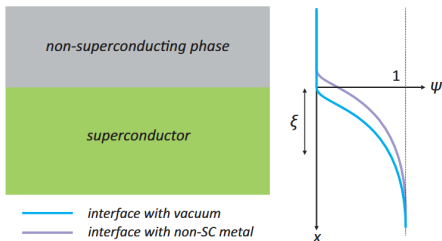
Ginzburg-Landau theory for superconductors(V)

Observations:

- **first GL equation** determines spatial behaviour of ψ .

Assume the situation in the following picture, one can compute from GL equation, with (def) **coherence length** $\xi = \sqrt{\frac{\hbar^2}{2m^*|\alpha|}}$

$$\psi(x) = \tanh\left(\frac{x - x_0}{\sqrt{2}\xi}\right)$$



Ginzburg-Landau theory for superconductors(VII)

How to **distinguish** between type I and type II superconductors?
Let σ be the surface energy, then the dimensionless ratio

$$\kappa = \frac{\lambda}{\xi}$$

is providing a tool

$$\kappa < 1/\sqrt{2}, \sigma > 0 \Rightarrow \text{type I}$$

$$\kappa > 1/\sqrt{2}, \sigma < 0 \Rightarrow \text{type II}$$

3. A framework for Numerical modelling of Ginzburg–Landau equations

Numerical simulation of vortex dynamics in Ginzburg-Landau-Schrödinger equation

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The rich dynamics of quantized vortices governed by the Ginzburg-Landau-Schrödinger equation (GLSE) is an interesting problem studied in many application fields. Although recent mathematical analysis and numerical simulations have led to a much better understanding of such dynamics, many important questions remain open. In this article, we consider numerical simulations of the GLSE in two dimensions with non-zero far-field conditions. Using two-dimensional polar coordinates, transversely highly oscillating far-field conditions can be efficiently resolved in the phase space, thus giving rise to an unconditionally stable, efficient and accurate time-splitting method for the problem under consideration. This method is also time reversible for the case of the non-linear Schrödinger equation. By applying this numerical method to the GLSE, we obtain some conclusive experimental findings on issues such as the stability of quantized vortex, interaction of two vortices, dynamics of the quantized vortex lattice and the motion of vortex with an inhomogeneous external potential. Discussions on these simulation results and the recent theoretical studies are made to provide further understanding of the vortex stability and vortex dynamics described by the GLSE.

Ginzburg-Landau-Schrödinger Equation (GLSE)

The time-dependent GLSE models vortex dynamics in superconductors, it reduces to GLE when $\alpha = 1$ and $\beta = 0$

$$(\alpha - i\beta)\partial_t\psi(x, t) = \Delta\psi + \frac{1}{\varepsilon^2} \left(V_0(x) - |\psi|^2 \right) \psi, \quad x \in \Omega^2, \quad t > 0,$$

Vortices occur where $\psi = 0$ with phase winding $m \in \mathbb{Z}$.

Phase winding is defined as

$$m = \frac{1}{2\pi} \oint_{\gamma} d\theta = \frac{1}{2\pi i} \oint_{\gamma} \frac{\nabla\psi}{\psi} \cdot d\mathbf{l}$$

The aim is to study the dynamics of these vortices.

Such a method works not only for superconducting materials.

Model Assumptions

Two spatial dimensions: $x \in \mathbb{R}^2$ or bounded domain.

Far-field condition: $|\psi| \rightarrow 1$ as $|x| \rightarrow \infty$ (bulk superconductivity).

Single-valued phase: vortices have integer topological charge m .

No magnetic field: pure scalar GLSE (zero vector potential $A = 0$).

Stationary far-field equilibrium: $\psi_\infty = 1$ phase reference.

Neglects thermal noise and dissipation.

1. Boundary conditions: $\psi(x, t) = e^{im\theta}$, $x \in \Gamma = \partial\Omega_R$, $t \geq 0$,

$$\psi(x, 0) = \psi_0(x), \quad x \in \overline{\Omega}_R$$

with Ω_R disk sufficiently large.

2. Take and decompose

$$(\alpha - i\beta)\partial_t\psi(x, t) = \Delta\psi + \frac{1}{\varepsilon^2} \left(V(x, t) - |\psi|^2 \right) \psi =: A(\psi) + B(\psi)$$

Time splitting:

- 1 Solve for Δt : $(\alpha - i\beta)\partial_t\psi(\mathbf{x}, t) = \Delta\psi$.
- 2 Solve for Δt : $(\alpha - i\beta)\partial_t\psi(\mathbf{x}, t) = \frac{1}{\varepsilon^2} (V(\mathbf{x}, t) - |\psi|^2) \psi$

In the particular case $V(\mathbf{x}, t) = V_0(\mathbf{x})$, there are exact analytical solutions.

Polar coordinates are used because of the geometry of solution (the vortices).

Results (I)

Two vortices with $m = 1$ each,
initial separation $d \rightarrow$

repulsion

Velocity is logarithmic:

$$v \propto \frac{1}{d} \log(1/d).$$

Escape to boundary unless
trapped by $V(x)$.

No annihilation.

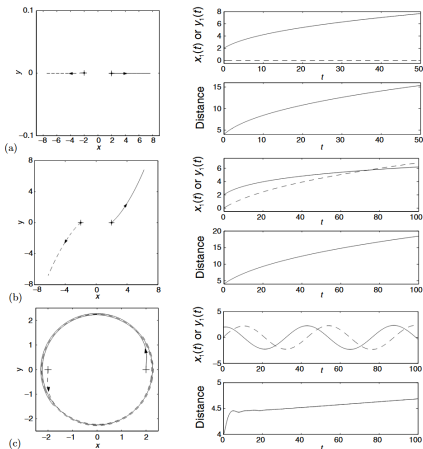
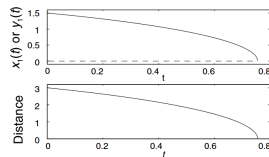
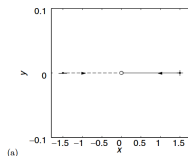


FIGURE 8. Time evolution of vortex centres for Case I with $a = 2$. (a) GLE; (b) CGLE; (c) NLSE.

Results(II)

One $m = +1$, one $m = -1$,
initial separation $d \rightarrow$
attraction.

Accelerate as d decreases.
Annihilate at center: cores
merge, $\psi \rightarrow 1$ everywhere.



Results(III)

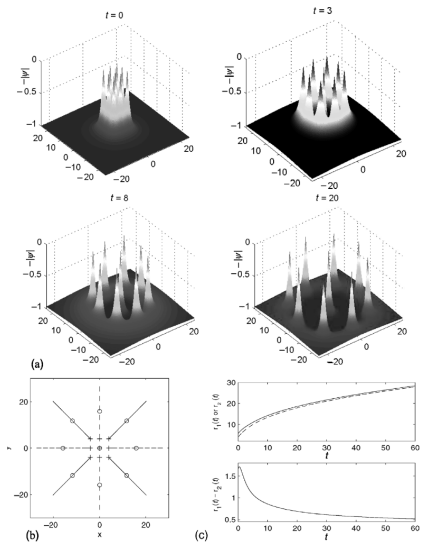


FIGURE 11. (a) Surface plots of $-\lvert\psi\rvert$ at different times; (b) trajectory of the vortex centres ('+' : $t = 0$ and 'o' : $t = 15$); and (c) time evolution of vortex centres, for dynamics of a vortex lattice with nine vortices (Case I) for the GLE.

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